Antecedents and Consequences of Personal Attachment in Cross-cultural Cooperative Ventures

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This study examines how personal attachments between boundary spanners within cross-cultural international cooperative ventures (ICVs) are established and their association with venture performance. Results of analysis of 282 ICVs in an emerging market (People's Republic of China) show that the development of personal attachment depends on factors at three levels. At the individual level, attachment is an increasing function of overlap in tenure between boundary spanners. At the organizational level, attachment is heightened by goal congruity between the parent firms but is impeded by cultural distance. At the environmental level, market disturbance and regulatory deterrence lead to strong attachments. Such attachments stimulate an ICV's process performance and increase financial returns.•

Relationships between partners are a prominent issue in all cooperative alliances. The dynamics of these relationships become even more fundamental in a cross-cultural setting, in which international cooperative ventures encounter more opportunities (e.g., mutual learning, knowledge transfer, market entry) as well as greater challenges (e.g., institutional volatility, cultural barriers, property rights protection) than do domestic ones. Since ongoing relationships between partners often become overlaid with social content (Granovetter, 1985), personal attachment between boundary-spanning managers becomes a critical force suppressing opportunism, boosting trust, and countering dissolution (Inkpen and Beamish, 1997).

Personal attachment reflects the socialization of boundary spanners during their involvement in continuous exchanges between the same interacting organizations (Seabright, Levinthal, and Fichman, 1992). Despite the importance of attachment, however, we do not yet have an adequate understanding of the ways in which personal attachment is established and how it then affects the performance of international cooperative ventures (ICVs). Most studies examining the effect of managerial ties have taken an external perspective on a domestic setting, that is, how interpersonal ties with managers or board members of other domestic firms, such as buyers, suppliers, and distributors, influence operational outcomes (Geletkanycz and Hambrick, 1997; Nohria and Eccles, 1992), diffusion of innovation (Heide and Miner, 1992; Westphal and Zajac, 1997), or alliance formation (Fichman and Levinthal, 1991; Gulati and Westphal, 1999). A significant remaining concern is how personal attachment between boundary spanners *within* an international cooperative alliance is established and how important this attachment is to improving joint payoff. This study aims to explore under what conditions personal attachment will be highlighted or undermined and how attachment affects outcomes of interorganizational relationships in the context of ICVs. Zaheer, McEvily, and Perrone (1998) demonstrated that interpersonal trust is positively linked to interorganizational trust but does not have a main effect on domestic supplier performance. This paper extends this research by shifting the emphasis from trust to attachment and from a domestic buyer-supplier setting to a cross-cultural joint venture. Attachment differs from trust in that attachment reflects the status

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quo of social connections between parties, whereas trust concerns the intention to accept vulnerability (risk) based on positive expectations of the behavior of another party. Reliance and risk are two necessary conditions for trust but not for attachment.

ICVs typically entail a considerable outlay of resources, create enduring and irreversible commitments between partners, and parties to them have difficulty controlling opportunism through formal governance mechanisms (Kogut, 1988). These characteristics make the role of personal attachment particularly crucial. While previous work on managerial ties and attachment is also informative for ICVs, ICV partnerships differ from domestic auditor-client, advertiser-client, or buyer-seller links in many respects, such as the external environment, nature of cooperation, degree of dependence, division of payoffs, resource contribution, and economic motivation, among others. Although Inkpen and Beamish (1997) postulated that lack of attachment is linked to ICV instability, the literature on ICVs has not yet examined how personal attachment is established, what factors influence the development of personal attachment, and how strongly attachment may influence ICV performance. This study seeks to fill this gap by examining these issues with data from ICVs in the People's Republic of China, an emerging market in which market mechanisms are incompletely developed. Economic theorists suggest that socially embedded relationships are invisible assets that distort the equilibrium of market mechanisms in a completely developed market and impede the well-being of stakeholders or society (Becker, 1976). When market mechanisms are incomplete, however, social ties substitute for a strong market structure and reduce organizational vulnerability to a weak structure (Williamson, 1985). As the world's largest emerging market, China has become host to the second largest number of ICVs, after the United States, and it should therefore be an ideal setting in which to examine the effects of personal attachment.

PERSONAL ATTACHMENT IN AN ICV

Personal attachment accrues over the period of time particular organizational boundary spanners are involved in exchange activities. In the context of an ICV, personal attachment represents the degree to which boundary spanners (resident board members and senior venture managers) from each party are socially bound through having developed interpersonal relationships and interpersonal learning. Without a personal relationship, interaction between boundary spanners is an economic rather than a social process. It is hard to sustain an economic exchange process for long because of moral hazards (i.e., opportunistic behavior of one party after the other party has already committed to the exchange) or uncertainty about individual and joint payoffs when two parties act simultaneously. Without interpersonal learning of complementary knowledge, boundary spanners lack an economic foundation to continue to interact.

When personal attachment within an ICV is established through socialization and reciprocal learning, it becomes social capital that can create economic value for the ICV.

Personal Attachment

Unlike economic capital, in which money or commodities are incorporated into cycles of production to create more profits, social capital does not necessarily involve making a deliberate investment to obtain a return. Instead, it is an aggregate of actual or potential resources arising from social interactions in the course of developing an institutionalized relationship.

Economic sociologists have articulated the importance of personal attachment in promoting long-term economic exchanges. Personal attachment serves as a conduit for information about administrative innovation between boundary spanners and helps to resolve uncertainty among top decision makers (Gulati and Westphal, 1999). Attachment enhances the trustworthiness of information circulating between boundary spanners and spurs the diffusion of innovations among connected members (Geletkanycz and Hambrick, 1997). Boundary spanners make substantial investments in such attachments to improve the effectiveness of the exchange relationship. An important attribute of such investments is the extent to which they are unique to a particular exchange relationship. According to Granovetter (1985), attachment is strengthened over time through personal contact as two parties develop specialized expertise peculiar to an organization's needs. Since specialized investments lose their value when applied to other relationships, the parties become locked into the existing relationship, which furthers continuity. Economic actions are thus embedded in the structure of social relations.

Knowledge exchange is central to the success of interorganizational cooperation (Inkpen and Beamish, 1997). Without personal attachment, the evolutionary process of knowledge exchange in long-term relational contracts such as joint ventures is difficult to complete. Personal attachment increases the willingness to transfer knowledge between boundary spanners. Increased trust derived from attachment makes boundary spanners more willing to transfer knowledge needed by their counterparts. The social structure of personal attachment within which economic actors are embedded also makes these actors realize that knowledge sharing is a reciprocal process imperative to sustaining individual ties (Powell, 1990). Personal attachment also improves the infrastructure for knowledge exchange between boundary spanners. As a source of enduring commitment from each party over time, personal attachment helps create a repository of reliable information on both the future needs for and the current strengths of knowledge from each partner. Greater understanding of the nuances of each other's knowledge, along with increased familiarity with the methods of communication that are effective, accentuate the ability to transfer knowledge (Zaheer and Venkatraman, 1995). Finally, personal attachments smooth out the process of knowledge exchange by facilitating the transformation of interpersonal links into interorganizational links. These micro-macro links are important in knowledge sharing because this knowledge is often embedded in the organization or group. Personal attachment creates a stable context within which interorganizational trust is developed and operational routines are institutionalized (Zaheer, McEvily, and Perrone, 1998).

The embeddedness and the importance of personal attachment are amplified in cross-cultural cooperation. In the long-term developmental process of ICVs, interorganizational relations evolve from information exchanges to more institutionalized relationships (Ring and Van de Ven, 1994). Socially embedded personal attachments within an ICV can be a counterforce to the liabilities of foreignness and to environmental threats (Boisot and Child, 1996). In this context, firms have greater difficulty obtaining information about the competencies and needs of their potential partners. Such information is often confidential and may not be revealed outside of a close relationship. Personal attachments thus foster mutual understanding between partners and reduce ambiguity in information. Moreover, firms entering ICVs face considerable moral hazards because of the unpredictability of their partners and the potential costs to an organization when a partner exhibits opportunistic or antagonistic behavior (Buckley and Casson, 1988; Kogut, 1988). In this situation, established personal attachments increase communication, cooperation, and trust. Such effects are further reinforced in emerging markets because firms there have a long tradition of doing business on the basis of interpersonal relationships. Personal attachment is not a given, however, but is a product of investment strategies aimed at establishing or reproducing social ties that will be directly useful over the short or long term (Burt, 1997; Inkpen and Beamish, 1997). As such, personal attachment is an endogenous factor, and its strength hinges on several other, antecedent factors.

Antecedents of Personal Attachment

Economic sociologists claim that relational exchanges involve social as well as economic exchange, both being developmental processes shaped by social norms and economic needs (Granovetter, 1985; Powell, 1990). Social embeddedness is influenced by individual traits as well as by the sociocultural environment (Boisot, 1986), while economic needs for governing ICVs are affected by interorganizational links and market disturbance (Parkhe, 1991). The level of personal attachment may thus depend on individual, organizational, and environmental factors simultaneously. This argument is akin to the tenet in social exchange theory that interpersonal links between boundary spanners are accumulated jointly by binding forces between individuals, between organizations, and within the society (Emerson, 1976; Ouchi, 1980). Forces at these various levels collectively shape trustworthy behavior or the possibility of ostracism in the evolutionary process of creating repeated personal ties (Cook and Emerson, 1978).

In a cross-cultural setting, individual attachment does not arise spontaneously but is deliberately developed through a continuous commitment to nurturing interorganizational exchange (Inkpen and Beamish, 1997). When there are cultural differences between boundary spanners, building interpersonal ties depends on knowledge about respective social norms, cultural standards, individual preferences, and skills. At the same time, an individual's commitment and behavior in cultivating personal attachment is shaped by the strategic needs of the organization he or she represents. When organizational needs and objectives differ substantially between two firms, personal attachment will be hindered. It is clear in theory on alliances that interorganizational cooperation through structural or personal attachments relies on congruence in partners' goals and cultural backgrounds (Park and Ungson, 1997). Additionally, ICV activities in an incompletely developed market are frequently disturbed by market changes and governmental interferences. As a counterforce to external changes, especially institutional hazards, personal attachment is used to reduce organizational dependence on the external environment and thus decrease associated transaction costs.

The antecedents of personal attachment thus encompass individual attributes that affect a boundary spanner's trustworthiness, organizational factors that affect congruence between partners, and environmental factors that affect transaction costs. Specifically, because the tenure of boundary spanners is associated with their level of familiarity with each other, while their educational background is linked to interpersonal learning, tenure overlap and education differences between boundary spanners are both likely to influence personal attachment. The tenure overlap between boundary spanners determines the amount of time during which personal attachment is established. Education differences predict asymmetry in boundary spanners' absorption capacity and in their ability to create synergy through interpersonal learning. When the strategic goals underlying ICV formation are congruent between partner firms, or cultural distance between them is lessened, boundary spanners will be able to work together in an environment that encourages personal attachment and individual commitment.

Individual determinants. Attachment is the result of an evolving history of personal ties. Developing personal relations depends on mutual understanding, information exchange, and trust building between boundary spanners, which are all an increasing function of their overlap in tenure. A high turnover of boundary-spanning managers can lead to loss of relationship continuity and to reduction of individual attachments (Seabright, Levinthal, and Fichman, 1992). Specific individuals constitute the repository of assets created from relationship-specific capital. As the overlap in tenure between foreign and local boundary spanners (i.e., how many years they have worked together within the venture in their boundary-spanning positions) increases, they gradually pass through a critical period of conflict and create a fruitful environment for developing relationship-specific assets. These assets may be in the form of skills embodied in people or institutionalized in organizational routines, assets, and technologies within an ICV (Martin, Swaminathan, and Mitchell, 1998).

The curve of interpersonal learning also depends on the duration of cooperation between boundary spanners. Boundary-spanning managers will not transfer, unilaterally or bilaterally, individually embedded tacit knowledge and skills to counterparts unless a requisite trust level has been established over time (Fichman and Levinthal, 1991). Trust between boundary spanners entails expectations of reliability and competence in the delivery of expected outcomes. When they establish trust over time, their continued exchanges will be facilitated even in the absence of legal and contractual mechanisms for monitoring their veracity. In an emerging market such as China, in which rules of law have not been well developed and laws are not strictly enforced, trust building between foreign and local managers takes time and requires tenure overlap but considerably heightens personal attachment once established. These arguments lead to the following hypothesis:

H1: The greater the tenure overlap between boundary spanners, the greater the personal attachment between them.

The educational level of a boundary spanner reflects the potential knowledge he or she can offer to another boundary spanner as well as his or her ability to learn and absorb skills from another boundary spanner. The smaller the educational differences (in terms of years of schooling) between boundary spanners, the higher the probability that they will be able to share knowledge with one another, which will nurture the development of attachment. By contrast, significant educational differences between boundary spanners temper knowledge sharing and reciprocal learning, which can hamper the development of attachment. Smaller educational differences also lead to a better common ground for communication, an important condition for advancing attachment.

The above effects are even stronger in a cross-cultural setting, although most foreign and local boundary spanners in ICVs have college degrees. Smaller educational differences between foreign and local boundary spanners may reduce individual cultural distances between them, as small distances imply that boundary spanners may have a more global vision and may be more receptive to innovative ideas and new managerial philosophies originating from different countries (Shenkar, 1995). Smaller educational differences also enable cross-cultural boundary spanners to share more common interests (e.g., ideals, goals, or hobbies), which is likely to strengthen interpersonal ties between them. Furthermore, a similar level or background of education is conducive to cooperation between boundary spanners and conflict reduction, thus intensifying attachment. In contrast, large differences in education are likely to plague cross-cultural communication, knowledge exchange, interest sharing, and cooperation between foreign and local boundary spanners:

H2: The greater the educational difference between boundary spanners, the lower the personal attachment between them.

Organizational determinants. Building attachment depends on the opportunity or potential for developing relation-specific assets (Levinthal and Fichman, 1988), which in turn depends on an atmosphere of cooperation between partners. The congruity of the goals set by an ICV's parents before it commences operations affects the extent to which partner firms will behave cooperatively or opportunistically during operations (Hill, 1990; Parkhe, 1991). Goal congruity harmonizes the interests of parties that would otherwise give way to the antagonistic pursuit of divergent goals (Williamson, 1985). As an agent of a parent firm, each boundary spanner accumulates personal attachment within the limits of governance and contractual frameworks stipulated ex ante between partner firms. Goal congruity between partner firms makes such frameworks more conducive to developing personal attachment.

When the strategic goals of each party diverge, boundary spanners are more likely to use distributive rather than cooperative strategies during ICV operations (Provan and Skinner, 1989), which is particularly harmful to building trust in a highly complex and uncertain environment. In contrast, goal congruence reduces each boundary spanner's perceived uncertainty about what other players will do, which in turn facilitates choosing the best responses to the predicted strategies of other parties (Gibbons, 1992). Moreover, goal congruity stimulates commitment from both parent firms and boundary spanners because commitment is necessary for ensuring both joint and private payoffs. This further encourages the development of personal relations and knowledge exchange between boundary spanners. Attachment is hence likely to be an increasing function of the congruity of the goals of the partner firms underlying ICV establishment:

H3: The higher the goal congruity between ICV partner firms, the greater the personal attachment between boundary spanners.

Cultural distance between partner firms is a particularly potent form of internal uncertainty for ICVs (Kogut and Singh, 1988) because it implies differences in managerial values, mind-sets, and norms (Hofstede, 1984). It may also lead to cultural ambiguity and process losses when boundary spanners from different cultures work together. Group homogeneity theorists hold that having culturally similar group members prevents losses in coordination, information flow, and communication within organizations (Shenkar, 1995). A high cultural distance between partners is likely to temper the establishment of personal relations between their boundary spanners and thus impede learning and knowledge sharing between them. Transferring competencies and capabilities is more difficult when cultural barriers are greater (Kogut and Singh, 1988). People whose cultural backgrounds are very different find it hard to communicate, share their experience, and verify each other's credibility (Carroll and Teo, 1996). Therefore, sharing skills between boundary spanners may be problematic when cultural distance is high. Finally, the ability to build personal relations is itself influenced by cultural distance. Knowledge of or skill at cultivating and developing personal relations is a form of cultural capital (Burt, 1997) that is embodied in those who know how to build, strengthen, and maintain interpersonal connections. Cultural capital can be gained individually through the process of learning, adaptation, and cultivation. Cultural distance interferes with creating this capital. The discussion above leads to the following hypothesis:

H4: The greater the cultural distance between ICV partners, the lower the personal attachment between boundary spanners.

Environmental determinants. One of the environmental factors that can affect attachment between boundary spanners in an ICV is market disturbance, the extent to which the industrial environment and related market structure are volatile and variable. Because market disturbance varies from industry to industry, each has different needs for and infrastructure for creating attachment. For instance, when extending Levinthal and Fichman's (1988) model on interparty attachment from the accounting industry to the advertising industry, Baker, Faulkner, and Fisher (1998) found a stronger effect of attachment on reducing dissolution in the advertising industry than in the accounting industry. Idiosyncrasies in market uncertainty and regulatory stringency in different industries are the underlying reasons for such differences. Unlike contractual risks resulting from the exposure of transaction-specific assets, which can be mitigated by internalizing intermediate markets, the uncertainty and risks of a volatile market are usually difficult to control (Williamson, 1985). When boundary spanners confront this hazard jointly, rather than separately, they can be either more cooperative (thus strengthening attachment) or more opportunistic (thus weakening attachment).

ICV managers in an emerging market such as China are likely to be more cooperative than opportunistic in this situation for three reasons. First, market opportunity is often accompanied by variability in an emerging market. Long stifled by ideology-based governmental control, market demand surges after the economy is decentralized and the market is opened. When both parties have more confidence in the market's potential and opportunity, they tend to be more cooperative in seizing market opportunities and attenuating market challenges. Second, although most ICVs have operated in these markets for just a few years, they are seeking longer-term economic benefits there. Market disturbances are often viewed as part of the necessary costs and risks of doing business there. Foreign and local boundary spanners are also likely to have a similar understanding of the nature and consequence of market uncertainty, which reduces possible conflicts and fosters cooperation between them. Third, both foreign and local managers find interdependency more important when facing a more volatile environment. Foreign boundary spanners must rely on their local counterparts' established ties with indigenous community and governmental authorities to reduce the ICV's vulnerability to external threats and to make it possible to acquire local resources. Local boundary spanners need to use the innovative skills and organizational experience of their foreign colleagues in jointly competing with rival firms and escalating the ICV's bargaining power with the local government.

The notion that market disturbance promotes conflict and thus limits attachment can hold true if one or both parties lose confidence in the potential of their collaboration or comprehend the consequences of market disturbance differently, but that is unlikely in an emerging market. Market potential is often accompanied by market disturbance when an emerging market undergoes structural transformation. When both parties view market disturbance as an inevitable side-effect of jointly benefiting from greater market opportunities, such disturbance is unlikely to result in new conflicts. If conflicts do arise, prespecified contractual guidelines about dealing with various contingencies arising from such disturbances can help boundary spanners overcome these conflicts. Moreover, unlike other market parameters, such as market demand and rivalry threat, which vary greatly across regions, industries, products, and periods of time within an emerging market, market uncertainty is common in all locations and sectors during economic transition and is thus unlikely to be understood idiosyncratically by different boundary spanners. The arguments above lead to the following:

H5: The greater the market disturbance facing an ICV, the greater the personal attachment between boundary spanners.

Another external challenge in an emerging market is regulatory deterrence, the extent to which business operations are hampered and deterred by administrative regulations imposed by a host government. Unlike government intervention, which may or may not reduce market efficiency, depending on market structure (e.g., monopolistic, oligopolistic, or competitive), regulatory deterrence prevents the market from becoming structurally complete and perfectly efficient by obstructing the movement of market forces, resulting in deviations from market equilibrium. Encountering such hazards, foreign and local boundary spanners are more likely to improve attachment than to increase internal conflicts. The likelihood of increasing conflicts seems slim because regulatory stringency in a transition economy is a common obstacle to all foreign-related businesses and is therefore anticipated by both foreign and local managers. Interparty negotiations about ICV formation inevitably have to deal with, and prepare for, jointly overcoming regulatory hazards. Mitigating such hazards is a common goal for both foreign and local boundary spanners because their decision making necessitates a stable and transparent institutional environment.

Enhanced attachment between boundary spanners can reduce the negative effect of regulatory deterrence on ICV operations. Scholars in the ICV literature have found that a foreign party's technologies and skills can enhance an ICV's bargaining power with the local government, while a local party's networks with indigenous officials can attenuate regulatory deterrence (Yan and Gray, 1994). When personal attachment increases, both parties will cooperate better and thus make better use of their respective competencies to buffer the ICV from regulatory deterrence. Studies in organizational theory suggest that regulatory stringency increases the impediments of the institutional environment and heightens the costs of obtaining, scanning, interpreting, and analyzing information concerning rules and policies (Duncan, 1972; Oliver, 1997). In ICVs, personal attachment is a cost-effective governance mechanism to stabilize internal operations and mitigate external hindrances. Using a local boundary spanner's knowledge, experience, and business contacts to cultivate a better relationship with governmental agencies is a more effective way to curb institutional hazards than simply trying to maintain distance from the government (Boisot and Child, 1996). But local managers are generally reluctant to share such knowledge or personal contacts with foreign counterparts unless they have built the necessary attachment. In Chinese society, personal connections are transferable to a third party only when strong individual attachment has been established, at which time personal connections become internalized and attachment becomes socially embedded. In an emerging market, where personal connections with officials are often more important than legal standards or impartial justice systems (Peng and Luo, 2000), greater regulatory deterrence pushes the ICV toward internal safeguards and joint efforts against external threats, which in turn heightens attachment. Thus, I hypothesize:

H6: The greater the regulatory deterrence facing an ICV, the greater the personal attachment between boundary spanners.

Consequences of Personal Attachment

The overlay of social relations in firms on what may begin as purely economic transactions plays a crucial role in the ultimate consequences of operations (Granovetter, 1985). Attachment helps maintain a relationship that, over time, reduces a firm's dependence on external resources or forces (Blau, 1964; Cook and Emerson, 1978). Opportunism and conflict are also curbed by personal attachment between boundary spanners. Attachment encourages trust building and enhances cooperation, and trust undergirds market activities and reduces vulnerability to external threats (Lewis and Weigert, 1985). In addition, the embeddedness of attachments generates standards of expected behavior that not only obviate the need for but are superior to pure authority relations in discouraging malfeasance (Granovetter, 1985). Personal relations are important in helping stimulate cooperation in economic life because norms of behavior deter opportunism, which is "curbed by the prospect of ostracism among peers" (Williamson, 1975: 106–108).

From a sociological view, personal attachment nurtures reciprocal forbearance (Cook and Emerson, 1978). A small amount of mutual forbearance can be transformed into a large amount of trust and cooperative efficiency within an ICV (Buckley and Casson, 1988). Ring and Van de Ven (1994) argued that personal bonds of friendship can lead to norms of group inclusion and increase the commitment of all parties to a cooperative relationship. A composite quasi-rent is often maximized in this environment (Hill, 1990). Personal relations between boundary spanners have a particularly strong influence on the reduction of opportunism in emerging markets like China's because opportunism can lead to loss of an individual's social "face," which potentially leads to loss of exchange opportunities with all members of that individual's network or even society overall. Because knowledge concerning the status of each individual's face travels quickly through an interpersonal network, an individual's loss of face can result in systemwide negative effects.

Personal attachment eases the settlement of disputes or conflict between parties to the ICV and thus contributes to its performance. The embeddedness of personal attachment often leads boundary spanners to negotiate solutions when problems arise that are not fully covered by the original contract. Such attachments attenuate the need to resort to contractual stipulations and legal sanctions (Williamson, 1985). Moreover, personal attachment mediates the convergence of managerial styles and cultural norms that were originally idiosyncratic to each party. Culture is not static but rather evolves over time as it is continuously constructed and reconstructed during interaction. A culture not only shapes its members but also is shaped by them, in part by their own strategies (Hofstede, 1984). As personal attachment accumulates, both parties develop a common culture that nourishes mutual learning and knowledge exchange, which further boosts the creation of operational and financial synergies. Lastly, personal attachment between boundary spanners is likely to lubricate interorganizational exchange. Individual attachment may facilitate organizational attachment by institutionalizing individually embedded experience and practices within ICV operations and management, which enhances ICV performance. Thus, I postulate:

H7: The greater the personal attachment between boundary spanners, the better the ICV performance.

METHOD

Data Sources

The hypotheses were tested using data from ICVs in China, which by the turn of the twenty-first century has become the world's largest emerging economy. China's per capita income has more than quadrupled since 1981, and real growth in gross domestic products has averaged 9 percent per year (World Bank, 2000). For a country whose population exceeds that of Sub-Saharan Africa and Latin America combined, this has been a remarkable development. Since 1979, when China opened up to foreign direct investment (FDI), through the end of 2000, about 400,000 foreign-funded enterprises (65 percent of them ICVs) representing U.S.$330 billion of FDI had actually been invested, accounting for about half of the worldwide FDI in developing countries and representing the second largest recipient of FDI in the world, surpassed only by the United States.

Data for this study came from three sources. Data on personal attachment and some proposed determinants and control variables were from surveys of ICV managers conducted in China in 1997 and 1998. Objective data on ICV performance came from a CD-ROM database available from the Ministry of Foreign Trade and Economic Cooperation (MOFTEC) in Beijing, China. This database contains key profile and performance data on each foreign-invested enterprise (FIE) that has operated in the country. Data on environmental determinants and cultural distance came from several archival sources, including the *China Statistical Yearbook.* Guided by theoretical considerations and field interviews, I developed a survey questionnaire in English, translated it into Chinese, and subjected it to a back-translation procedure. Both versions were initially cross-checked by three management professors in China. I then pretested the questionnaire's validity with 28 ICV managers representing 14 ICVs (paired respondents from each venture) in Jiangsu Province. I interviewed these managers, asking them to respond to all questionnaire items about levels of personal attachment and ICV performance within their respective ICVs. The results revealed a high intraclass consistency between the two paired respondents, both on attachment (intraclass r: .71–.89) and on performance (intraclass r: .62–.83). I also asked these managers to identify any ambiguities in terms, concepts, or issues in the questionnaire and made some minor changes of wording based on their feedback.

Sample firms were identified mainly from the *List of Foreign-Invested Enterprises in China,* published by China's State Statistical Bureau in 1994, and the *Directory of Foreign-Invested Enterprises,* compiled by MOFTEC in 1996. Only equity manufacturing joint ventures fall within the scope of the present study. Equity joint ventures account for about 75 percent of all cooperative ventures established recently in China. Nonequity joint ventures were not considered because most of them were not independent entities. An independent contractor in China sent questionnaires to a random sample of 800 ICVs, many of them in such provinces as Jiangsu, Zhejiang, Shanghai, and Shangdong, major regions that have been attracting foreign investments for many years.

The respondents were the ICVs' general managers (or, in 10 cases, deputy general managers) whose names were given in the above *List* and *Directory.* A phone call to each ICV determined whether those listed remained in the key managerial position of the venture. When a manager was no longer in the venture (35 out of 800 ICVs sampled), we sent the questionnaire to the new general manager whose name was identified through phone contacts. In addition, the questionnaire asked each respondent to indicate his or her position in the ICV. Those informants who were not general managers or deputy general managers were omitted from the sample. After several rounds of follow-up reminders, 282 useful questionnaires were received, representing a 35.25 percent response rate. These were from ICVs mostly formed before 1990, engaged in various industries, and having partners from diverse countries of origin. Of the total respondents, 97 were foreign expatriates and 185 were local nationals. The average number of employees was 680, and the average total investment by all parties was $14.2 million.

To provide a triangulation with some of the mail survey results and between respondents within the same company, semistructured interviews were conducted with 44 senior managers, representing 22 ICVs in Jiangsu Province, right after the completed questionnaires were received. These managers were identified from code numbers we had stamped on each questionnaire. They were asked to respond orally to questions raised in the questionnaire. The results exhibited a high consistency between their reports and the answers in the questionnaire (Pearson correlations: .49–.77) and between the two respondents from each firm (Guttman R for attachment = .87, for performance = .80, and for other survey items = .82–.88).

Using the database created by the Computer Center at MOFTEC, I also checked for nonresponse bias. I was able to obtain archival data on some ICV characteristics, namely, total investment, foreign equity, duration, sales, and profit for both responding and nonresponding ICVs in the sample. I tested for differences between the responding and nonresponding ICVs on these characteristics using a multivariate discriminant analysis. Hotelling-Lawley's T was nonsignificant between the two groups (T = 0.08, F = 1.13, $p > .10$), suggesting that there was no bias from nonresponses.

Variables

Personal attachment was measured by the average of the responses, on a 7-point Likert scale, to the following items: (1) the extent to which interpersonal relations have been developed to date between each party's boundary spanners since the ICV was formed; (2) the extent to which interpersonal relations had been developed between these boundary spanners through previous trade, investments, or negotiations before the ICV was formed; (3) the extent to which foreign boundary spanners have provided Chinese boundary spanners with needed personal skills and knowledge; and (4) the extent to which Chinese boundary spanners have provided foreign boundary spanners with needed personal skills and knowledge. In each question, "boundary spanners" were defined as the ICV's resident board members and senior managers (general or deputy general managers) from each party. Each respondent was asked to represent the venture in answering the above questions in his or her capacity as an ICV's leading manager. As the effects of some of these items multiply in interaction, I averaged them using a geometric instead of an arithmetic approach.[1] The high Cronbach's alpha (.73) and communality estimates (.85–.93) from the exploratory factor analysis confirmed the internal consistency and dimensionality of this construct.

Antecedent variables. The length of *tenure overlap* between boundary spanners was measured by the arithmetic average of tenure overlap (1) between two parties' resident board members and (2) between two parties' top managers (general or deputy general managers). This overlap was defined as the number of years working together in the venture with a counterpart in a similar position. A respondent from each ICV was asked to report this tenure overlap between these boundary spanners in his or her ICV. The *education difference* between boundary spanners was measured by the difference between the average number of years of schooling of foreign boundary spanners and that of local boundary spanners, as reported by one respondent per ICV. *Goal congruity* was measured by asking each respondent, on a 5-point Likert scale, to assess the overall congruity of the strategic goals behind the establishment of the ICV set by foreign and local parents before the commencement of ICV operations. Its content validity was validated by a covariance structure analysis, which reported a significant path correlation ($p < .001$) between this congruity and the mean of specific goal differences (reverse coded) between parent firms (profitability, risk reduction, local market expansion, and knowledge sharing). Most of the respondents had participated in ICV negotiations and knew the goal differences between parent firms from these negotiations. Unlike the operational goals set by ICV managers, the strategic goals of parent firms concerning why

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I tested the significance of the interaction between these sub-items after calculating the arithmetic mean. Both correlation and regression tests suggested that the interaction effect is significant at $p < .05$ or lower in relation to most of the proposed antecedents and performance measures. Because each sub-item is insufficient to capture the essence of attachment, I thus defined attachment as a multi-item construct amounting to the geometric average of these sub-items.

they are establishing an ICV are unlikely to change after ICV formation.

The *cultural distance* between parties from different countries was computed following Kogut and Singh's composite index (1988) and based on the data from Hofstede (1984) and Huo and Randall (1991). This index measures deviation along each of four cultural dimensions (i.e., uncertainty avoidance, individuality, power distance, and masculinity-femininity) from the score of a given focal country for each foreign country j such that:

$$CD_{j/china} = 1/4\Sigma[(I_{ij} - I_{ichina})^2/V_i]$$

where I_{ij} = the index value for cultural dimension i of country j; I_{ichina} = the index value for the cultural dimension of China (PRC); and V_i = variance of the index dimension i. The name of the foreign country involved in each ICV was identified from MOFTEC's database. When the nationality of the key foreign boundary spanner (a resident board member or general manager) differed from that of the foreign parent, I used the boundary spanner's nationality (the information obtained from the survey) in computing the distance.[2]

Market disturbance was measured as the geometric average of standard deviations of five structural market attributes, including an industry's total sales, profit after tax, output, capital investment, and the number of firms between 1994 and 1997:

$$[\text{Std(sale)} \times \text{Std(profit)} \times \text{Std(output)} \times \text{Std (investment)} \times \text{Std(\# of firms)}]^{1/5}$$

This way of operationalizing structural disturbance is appropriate because standard deviation is a common measure of volatility, and market structure is a multidimensional construct. Using a geometric average instead of simply multiplying the standard deviations of the structural attributes prevents a skewed distribution. A plot test shows that this approach presents a more normal distribution on this variable than does taking logs. Information about the industry in which an ICV participates was obtained from MOFTEC's database, while the data regarding the five structural attributes came from the *China Statistical Yearbook* (1994–1997 editions).

Finally, information on *regulatory deterrence* was obtained from *The Orientation Directory of Industries for FDI,* compiled by the State Council, China, which explicitly states that governmental priorities and institutional support are given to industries as categorized by priority from 5 to 1. Detailed measures encouraging or restricting FDI, as stipulated in this directory, further manifest governmental support in this sequence. Thus, the higher the number, the greater the regulatory deterrence, and vice versa. This objective measurement of deterrence correlates highly (correlation coefficient:

2
This study did not tease out an effect on cultural distance of characteristics of managers at the parent firms who choose boundary spanners, although this effect is unlikely to be strong. Selection of boundary spanners in ICVs is often made by parties in proportion to their relative bargaining power.

.79) with overall regulatory deterrence as perceived by senior ICV managers (on a 5-point-scale item in the questionnaire describing the degree of deterrence to ICV operations). Using the directory's categories, regulatory deterrence was defined as 5 if an ICV's industry lies in Restricted Category B, 4 if in Restricted Category A, 3 if in the Newly Opened Category, 2 if in the Encouraged Non-Technological Category, and 1 if in the Encouraged Technological Category.

ICV performance. Performance was defined by two measures. The subjective construct, *process performance,* was the mean of managerial satisfaction, on a 5-point Likert scale, with ICV performance in (1) managing the venture, (2) developing technology, (3) product design, (4) quality control, (5) labor productivity, (6) marketing, (7) distribution, (8) customer service, (9) cost control, and (10) organizational reputation. The respondents were asked to refer to the recent three years as the time frame for answering these items. The communality estimates (from .84 to .91) from an exploratory factor analysis indicated that each of the above ten processes is an important area of process performance. Cronbach's alpha (.79) also validates the internal consistency of this construct. Arino (1996) showed that such subjective performance measures for ICVs have strong convergent and discriminant validity. The objective measure, *return on investment* (ROI), was calculated as the total profit before tax divided by total investments, using information from the MOFTEC database (1994–1996).[3]

In assessing the determinants and consequences of attachment, I controlled for *investment size* ($ million), *equity arrangement* (percent owned by foreign party), and *project location* (coastal vs. non-coastal area) at the time an ICV was set up. In addition, I controlled for *length of ICV operations* (number of years so far) in China and *industry sales growth* (compound sales growth of an ICV's industry). The degree of personal attachment and the level of ICV performance may vary according to these variables. Information about industrial sales growth was obtained from the *China Statistical Yearbook* (1994–1996), while all other control variables were measured using information from the MOFTEC database. Table 1 displays descriptive statistics of the above variables and Pearson correlations between them.

RESULTS

I used structural equation modeling (SEM) analysis to test the hypotheses. Overall, the model statistics (maximum likelihood estimation under covariance structure analysis) suggest a good fit of the proposed model with the data. As figure 1 shows, the model has a chi-square value of 96.94 ($p < .001$), together with above-threshold levels on four goodness-of-fit indices. Omitting personal attachment from the model significantly reduced the goodness of fit of the new model (GFI = .44, and CFI = .47). The Tucker-Lewis incremental fit index (TL = .98) further suggested that omitting personal attachment from the model significantly decreases overall model fit. This confirms that personal attachment has an overall mediating role in linking a set of antecedents of attachment and ICV performance: these antecedents influence attach-

3
Reverse causality may exist if ICV performance is defined as longevity or survival or as previous performance or expected future performance. The longer survival of an ICV might lead to higher attachment through trust building. High ex ante performance may increase attachment through reduced conflict. One would also expect ICV boundary spanners to become more attached when there is a greater expectation of future success. This study, however, is limited to examining how current performance is influenced by attachment accumulated in the past.

Table 1

Descriptive Statistics and Pearson Correlations (N = 282)

Variables	Mean	S.D.	1	2	3	4	5	6	7	8
1. Return on investment	0.14	0.25								
2. Process performance	3.50	0.65	.44•••							
3. Personal attachment	5.01	1.19	.18••	.28•••						
4. Tenure overlap	3.27	1.08	.11	.07	.27•••					
5. Education difference	4.11	1.60	.07	–.04	–.09	–.08				
6. Goal congruity	3.38	1.91	.16•	.17••	.23•••	–.11	.10			
7. Cultural distance	5.88	1.29	–.14•	–.06	–.26•••	.05	–.04	.06		
8. Market disturbance	7.27	3.66	–.33•••	.23•••	.25•••	–.08	–.07	.09	.01	
9. Regulatory deterrence	3.26	1.08	–.28•••	.40•••	.24•••	.02	.01	–.13•	.05	–.16••
10. Investment size	14.2	4.67	.03	.19••	–.21••	.10	–.09	.04	–.16•	–.08
11. Equity arrangement	0.48	0.33	.15•	.12	.12	–.05	.02	–.01	.05	.01
12. Project location	0.59	0.49	.10	.15•	–.04	.06	–.11	–.06	–.07	.05
13. Length of operations	8.67	1.45	.25•••	.25•••	.16••	–.06	–.08	–.05	–.07	–.17••
14. Industry sales	13.4	4.19	.19••	.12	.20•••	.04	.03	.05	–.11	–.02

Variables	9	10	11	12	13
10. Investment size	–.03				
11. Equity arrangement	–.21•••	–.09			
12. Project location	–.07	.08	–.07		
13. Length of operations	–.14•	.09	.10	.18••	
14. Industry sales	–.01	.10	.06	–.09	–.08

• $p < .05$; •• $p < .01$; ••• $p < .001$.

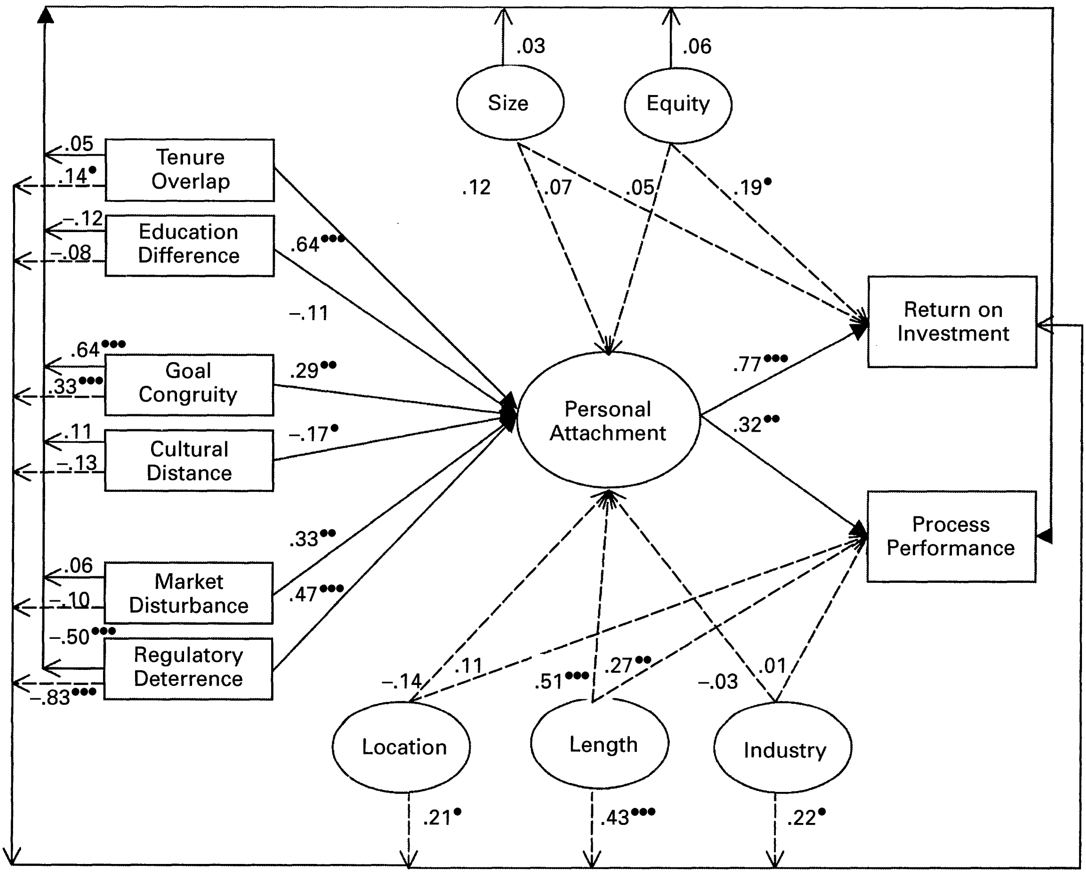
ment, which in turn influences performance. When I split the total sample randomly in half and reconducted the SEM, the major results (path directions and significance) remained unchanged from figure 1, suggesting that the results in figure 1 are not due to chance.

As figure 1 shows, tenure overlap is significantly and positively associated with personal attachment, suggesting that longer working relationships between boundary spanners lead to higher attachment. I also checked the causal effect of attachment on tenure overlap and found that this effect is nonsignificant ($\beta = .12$, $p > .10$). Thus, the tenure-overlap–attachment relationship is a one-directional link (longer tenure overlap leads to greater attachment) rather than a reciprocal link between them. Although education difference has a negative coefficient in relation to attachment, as hypothesized, it is not statistically significant. This evidence implies that attachment is not necessarily affected by the educational difference between boundary spanners in terms of years of schooling. It seems possible that less educated Chinese managers may use personal ties with other senior managers more intensively in order to solidify their managerial position within ICVs. This would corroborate Xin and Pearce's (1996) finding that less educated entrepreneurs in China appear to use more *guanxi* (interpersonal connections that imply a continued exchange of favors) to obviate internal or external threats and mitigate the effects of their own weaknesses. The negative effect of differences in education on interpersonal learning may thus combine with the effect of increased personal ties, which on balance makes the links between education difference and attachment nonsignificant.[4] The above findings thus support H1 but not H2.

The SEM analysis shown in figure 1 also suggests that goal congruity between parental firms has a positive influence on

4
When I split the sample based on an industry dummy (1 if technology/capital-intensive and 0 otherwise), the SEM analysis suggested that educational difference has a marginal effect on attachment ($\beta = -.16$, $p = .084$) for firms in technology/capital intensive industries, but there was no such effect in the group of non-technology/capital-intensive sectors. The link between educational difference and attachment thus varies between industries that require different levels of education and knowledge.

Figure 1. Path coefficients of structural equation modeling analysis of personal attachment (N = 282).*



Model evaluation with goodness-of-fit measures:
Chi-square: 96.94, $p < .001$
Goodness-of-fit index (GFI): .95
Bentler & Bonett's normed fit index (NFI): .92
Adjusted goodness-of-fit (AGFI): .908
Comparative fit index (CFI): .94

•$p < .05$; ••$p < .01$; •••$p < .001$.
* None of the path coefficients that reflect causal effects of control variables (size, equity, location, length, and industry) on antecedents of attachment are significant, nor are reverse causalities from these antecedents to control variables, except for a significant effect of regulatory deterrence on foreign equity ($\beta = -.36$, $p < .01$).

personal attachment, while cultural distance between them exerts a negative effect. I checked for reverse causality, and the result showed that attachment does not influence ex ante goal congruity ($\beta = .07$, $p > .10$). This seems logical because the strategic goals behind ICV formation are set by the parent firms. Boundary spanners, even strongly attached, seldom have the power to change the goals of their parent firms. The above finding supports the argument that boundary spanners are likely to use cooperative rather than distributive strategies when the strategic goals of the parties converge. In the language of game theory, a Nash equilibrium is reached when boundary spanners know what their counterparts will do and can make the best responses to achieve maximum joint payoffs. Such behaviors and outcomes are hindered, however, by the cultural distance between foreign and local partners. Thus, H3 and H4 are supported.

Two environmental antecedents, market disturbance and regulatory deterrence, are both positively linked to the level of personal attachment. This evidence lends support to H5 and H6. This result, however, may be China-specific or apply only to other similar emerging markets rather than all contexts. Most foreign companies entering this market are seeking the long-term economic benefits to be gained from preempting volatile opportunities and are willing to take the necessary risks to fulfill their strategic goals. This risk-taking behavior encourages them to use personal attachment to counter external hazards. Cooperation between partners arising from this attachment reduces an ICV's dependency on external resources and vulnerability to changes in these resources. In contexts in which abnormal rent-generating opportunities are rare, however, market disturbance and institutional hostility may reduce organizational or personal commitment to the partnership, thus decreasing the likelihood or rate of developing attachment.

Among the control variables, ICV location (1 if coastal cities, 0 otherwise) is negatively rather than positively associated with personal attachment. This suggests that boundary spanners in ICVs located in non-coastal cities tend to develop stronger personal attachment than those in ventures in coastal cities. ICVs in inland regions have fewer institutional privileges granted by the central government and operate in a less favorable investment environment than do their counterparts in coastal cities. Thus, ICV managers in inland regions may use stronger attachment to circumvent environmental disadvantages. Another control variable, the length of ICV operations in China during which attachment is developed, also shows a positive path coefficient with personal attachment. Thus, the longer the ICVs have been in China, the higher the personal attachment between boundary spanners. The two time-related variables, namely, individual tenure and length of operations, are therefore both important for developing personal attachment within an ICV. Other control variables do not have a sizable impact on attachment.

Figure 1 also presents results on the performance implications of personal attachment. Attachment is positively and significantly associated with ICV process performance and return on investment. This validates H7, which proposed that personal attachment would have a favorable influence on ICV performance. When interpersonal ties and interpersonal learning are separated from the attachment construct, both remain significant and positive in relation to process performance and return on investment. The additional path analysis also confirms that there is no reverse causality between process efficiency and attachment (β = .15, p > .10) and between ROI and attachment (β = .05, p > .10), suggesting a one-directional rather than a reciprocal link between attachment and performance measures. This seems logical, since the analysis examined how accumulated attachment is related to current performance. Although return on investment does not represent all facets of an ICV's ultimate outcome, the above results at least show that attachment between boundary spanners makes a strong contribution to profitability. More importantly, attachment contributes critically to process efficiency. This is important because process efficiency is a prerequisite for any kind of ICV success that the parent firms endeavor to accomplish.

Personal attachment is certainly not the sole factor affecting ICV performance in a dynamic market. The SEM analysis shows that several organizational and environmental factors have a strong, direct effect on both process outcome and return on investment. Results show that length of operations in a host country and goal congruity between investing firms both significantly enhance ICV performance. A larger equity share owned by a foreign party is also associated with higher return on investment. Additionally, some external factors, such as location, industry growth, and regulatory deterrence, are strongly related to return on investment. Market disturbance, however, does not affect firm performance. This can be explained by the fact that growth opportunity and market volatility coexist in many newly opened sectors. A variable and dynamic environment does not negate the likelihood of a high payoff.

DISCUSSION

The results of this study shed some light on how personal attachments between ICV boundary spanners are established and how such attachments are associated with ICV performance. By extending the extant literature on attachment to ICVs in an emerging market context, this study showed that the level of personal attachment is contingent on several factors at three levels: individual, organizational, and environmental. At the individual level, attachment is an increasing function of accumulated tenure overlap, that is, the length of time boundary spanners have been working together. At the organizational level, goal congruity between the parent firms spurs attachment between boundary spanners, whereas cultural distance between two firms impedes the development of attachment. At the environmental level, market disturbance and regulatory deterrence, the two most notable characteristics of emerging markets, lead to more rather than less personal attachment. More critically, this analysis demonstrates that personal attachment significantly contributes to process performance as well as return on investment for ICVs. Several control variables, such as location, equity, industry, and length of operations, also influence attachment or performance.

The above outcomes provide the basis for some theoretical and managerial insights into interorganizational exchange in a cross-cultural setting. The most distinctive aspect of the results is that personal attachment is both endogenous and exogenous. It is endogenous because attachment itself is not a given. Instead, it is determined by various factors that influence managerial ties and learning between boundary spanners. It is exogenous because personal attachment predicts variance of ICV performance. Personal attachment drives up both the process and ultimate (i.e., ROI) performance of ICVs. The structural inertia paradigm in organizational theory suggests that attachment constitutes a counterforce to change rather than a pressure for change (Stinchcombe, 1965). This view implies that ties between boundary spanners may impair process efficiency by creating managerial collusion, heightening agency costs, or slowing innovative and adaptive changes. The results of this study of a sample of ICVs in China, however, suggest that attachment has a positive role in creating a joint payoff, including improved process efficiency. Zaheer, McEvily, and Perrone (1998) demonstrated that personal trust does not have a main effect on domestic supplier performance in the United States. It appears, therefore, that the role of attachment or trust may differ in different organizational settings. In the context of an ICV in which both parties are bound by an equity commitment, attachment fosters cooperation and trust building, which in turn helps create joint venture synergies. Because attachment involves knowledge sharing, it seems logical that attachment may exert a stronger effect on performance than trust. The role of attachment is reinforced by environmental dynamics in an emerging market. When boundary spanners realize the importance of seizing opportunities or mitigating disturbance, their attachment becomes a powerful force that stimulates ICV growth. Development of theory on interfirm partnerships, especially those emphasizing trust and cooperation, may also benefit from these results: individual attachment is a catalyst for cooperative payoffs, and this attachment is transferable to interorganizational collaboration. Current research on ICV attachment has emphasized its effect on ICV stability. This study goes further, in demonstrating that attachment is fundamental to improving an ICV's process results as well as its financial returns.

Second, this paper has documented multilevel factors underlying the development of personal attachment. Transaction cost economists generally emphasize how market forces influence attachment, while social exchange theorists often focus on how individual or organizational traits affect attachment. Each of these views alone seems incomplete in deciphering the underlying determinants of personal attachment. This study shows that attachment hinges on multilevel factors related to individual, organizational, as well as environmental characteristics. Although the antecedents identified here may still be incomplete, the analysis demonstrates that they do not originate from a single level or source. This implies that personal attachment is not just a personal matter; it is also shaped by ventures' attributes, such as goal congruity and cultural distance, and by contextual dynamics, such as market disturbance and regulatory deterrence. This further implies that some antecedents (individual and organizational) are controllable while others (environmental) are not. Knowing about such antecedents is useful for multinational companies attempting to provide appropriate conditions for developing and stimulating personal attachment. For organizational theorists, this evidence is a reminder that the evolution of personal attachment is jointly determined by micro- and macro-level factors that involve economic logic and social exchange and that it proceeds dynamically in a complex process. Given that the nexus of an interparty relationship rests mainly on the attachment between boundary spanners, this analysis extends theory development in trust building by revealing how interpersonal attachment is developed in a cross-cultural and interorganizational context, set in a complex and unpredictable institutional environment.

Third, the logic of theories of interfirm cooperation has been dominated by economic or strategic explanations, such as resource dependence, transaction cost reduction, and interpartner learning. The significant impact of attachment on performance found in this study suggests the importance of social embeddedness in the evolution of ICVs. Economic or strategic perspectives do not provide a satisfactory account of social embeddedness. Social embeddedness arising from attachment may have implications beyond the persistence of a relationship. Although it is empirically difficult to distinguish between attachments that reflect transaction efficiency and those that reflect the embeddedness of a relationship, as the two properties are likely to covary, it is hard to imagine the development of highly specific relationship capital that does not engender some social obligations. As Granovetter (1985) suggested, embedded relationships should reduce opportunism and, therefore, transaction costs. Interorganizational exchange during the development of ICVs is an economic as well as a social phenomenon, simultaneously shaped by profit-driven incentives and socially binding forces. Personal attachment is predicated on both economic and social rationales. In a cross-cultural setting involving a transition market, the importance of embeddedness is magnified because relationship capital is sometimes even more important than economic capital in undertaking business transactions. This article provides evidence that individual attachment is an important source of competitive advantage that helps increase the likelihood of an ICV's success.

Fourth, the results of this study demonstrate that several individual (tenure overlap), organizational (goal congruity, length of operations, and location), and environmental (regulatory deterrence and industry growth) factors affect both attachment and performance. They imply that these variables affect venture performance independently as well as through the mediating channel of attachment. For instance, results showed that regulatory deterrence and goal congruity directly affect ICV performance while increasing personal attachment. Such findings support arguments by Williamson (1985) and others emphasizing the importance of personal attachment in modifying the linkage among other business factors and economic efficiency. Attachment is a developmental process that is not isolated from other exogenous variables that have consequences for transactions. This may enrich the theory of interfirm cooperation in establishing that individual attachment can buffer an ICV from the effects of internal opportunism or external hazards while spurring contributions to interfirm forbearance and cooperation. In addition, the results on ICV performance may offer some insights into the consequences of interfirm partnerships, an understudied issue in the field. They demonstrate, for instance, that institutional factors are particularly critical in shaping ICV performance. This is manifested not only in the importance of regulatory deterrence but also in the significance of venture location and industry growth. The role of location and industry in affecting performance largely rests in the variation in investment policies by regional governments and in the differences in operational environments across regions and industries.

Lastly, the analysis in this paper suggests that theories on personal attachment apply to the context of a foreign emerging market such as China. Such theories integrate the wisdom of both economic and sociological perspectives and hold true for analyzing the evolution and role of attachment in firms in a complex and uncertain environment. When an ICV faces external hazards while preempting market opportunities, its managers develop attachments and use them to make up for organizational weaknesses such as the liability of foreignness. Attachment saves transaction costs because it counters environmental uncertainty, reduces moral hazards, and provides benefits to the ICV from local managerial connections to the host business community. The finding of a positive effect of ownership equity (a proxy for governance level) on return on investment provides further evidence of transaction cost rationality. Without this economic basis, the sociological view cannot fully explain the systematic and sustained contribution of personal attachment to process performance and accounting returns for ICVs in this context. The positive influence of the length of ICV presence on both attachment and performance corroborates the notion that the development of both interpersonal relations and firm growth necessitate an investment of time in a social environment in which businesses are often conducted on the basis of personal connections. This study also joins others in emphasizing the effect of the environment on firm operations in China (Boisot and Child, 1996; Peng and Luo, 2000) and shows that regulatory deterrence, industrial dynamics, and project location all have significant impacts on firm performance.

The results of this study suggest several potentially fruitful avenues for future exploration. First, they provide an impetus for exploring the relationship between personal (individual) and structural (organizational) attachment and how they interactively influence ICV performance. Research in organizational behavior and sociology (especially on networks) has offered many insights into the influence of structural factors on individual attachment. Formalized procedures are likely to replace the coordination of interorganizational activities through personal contact. Relationship-specific skills can be embodied both in individuals and in routines that are institutionalized and divorced from particular individuals. What remains unclear is how structural attachment and personal attachment influence each other in international networks. Parties with different cultural backgrounds may undertake and configure these two attachments differently. Another research question is whether conditions in foreign environments modify such configurations.

The second avenue for research stems from this study's limitation of a cross-sectional design. To check whether there were threats to the validity of major findings due to this design, I split the total sample randomly in half and found that the results still held up. Further, I tested several alternative specifications (reverse causalities) in the structural equation model and found that those alternative specifications were not significant. These checks suggest that the results have reasonable validity. When one is concerned with how accumulated attachment is associated with evolving multilevel antecedents or with longitudinal performance, however, a cross-sectional design will be deficient in providing such longitudinal insights and assessing reverse or reciprocal causalities. Future research should investigate whether previous performance has a causal effect on current attachment and whether current attachment has a causal effect on future changes in goal congruity and on future tenure overlap. A longitudinal analysis is warranted and will enrich our understanding of the dynamic role of personal attachment in the developmental process of ICV growth.

A third important direction is exploring which individual attachments are important in relation to specific areas of performance for large, sophisticated ICVs. For instance, it would be interesting to see whether attachment between resident board members from two parties to the ICV has a stronger effect on overall performance than attachment between senior managers, which might exert a more critical influence on process performance. Moreover, a cross attachment between one party's resident board members and the other party's senior managers may also affect interorganizational exchange and its consequences. Decomposing individual attachments is necessary because different individual groups are responsible for and sensitive to varying aspects of operations and management. Similarly, disaggregating multiple dimensions of attachment (economic vs. social embeddedness or interpersonal learning vs. interpersonal ties) would be a useful effort because it could offer new insights into the structure of attachment. Determining which interpersonal attachments bear on which aspects of the relationship and performance should help provide a finer-grained picture of these relationships as well as clarify what the dynamics of these attachments are.

Finally, as a departure from this study, future research could investigate how asymmetry in bargaining power between partners influences the nature and persistence of personal and structural attachments and how the shifting bargaining power and the dynamism of attachment are linked. The evolution of an attachment may be biased toward the interests of the boundary spanner or party who has the dominant decision-making or bargaining power. This research question is important because power asymmetry and attachments may exert opposing forces on the stability and longevity of the ICV. While asymmetry constitutes a source of ICV instability, attachment may counter this instability. Thus, knowledge about an interrelationship between asymmetry and attachment would cast new light on the combined effect of asymmetry and attachment and help predict which force dominates in determining the stability and persistence of an ICV.

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